

Deepfake Video Authentication using Explainable Temporal AI

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Abstract—The rapid advancement of generative artificial intelligence has enabled the creation of highly realistic manipulated videos, commonly referred to as deepfakes. Although existing deepfake detection systems have achieved promising performance, many approaches primarily focus on frame-level visual artifacts and produce a binary real/fake decision without providing sufficient temporal or forensic evidence. This work proposes an explainable spatial-temporal framework for deepfake video authentication that analyzes both facial appearance and temporal consistency across consecutive video frames. The proposed framework uses face preprocessing and alignment, Xception-based spatial feature extraction, Temporal Transformer-based sequence learning, spatial-temporal feature fusion, deepfake classification, temporal suspicion analysis, and Explainable Artificial Intelligence. LSTM and GRU models are retained as comparative temporal baselines, while the Temporal Transformer serves as the primary temporal architecture of the proposed system. Grad-CAM and attention-based visualization are incorporated to identify facial regions and temporal positions that influence model predictions. In addition, the framework is designed to estimate suspicious temporal segments rather than relying exclusively on a video-level classification. The system can also report visual instability and blur/clarity fluctuations as model-derived temporal evidence where supported by the feature analysis. An interpretable authenticity score is proposed using calibrated model probabilities. The framework will be evaluated on publicly available deepfake datasets using accuracy, precision, recall, F1-score, ROC-AUC, and, where temporal annotations are available, segment-level localization metrics. The primary research objective is to investigate whether integrating spatial analysis, temporal consistency modeling, explainability, temporal suspicion localization, and calibrated authenticity scoring can provide a more informative and forensic-oriented deepfake authentication workflow than conventional frame-centric classification.

Index Terms—Deepfake Detection, Video Forensics, Temporal Analysis, Explainable AI, Grad-CAM, Computer Vision, Deep Learning, Facial Analysis, Video Authentication

I. INTRODUCTION

The increasing accessibility of generative artificial intelligence has significantly reduced the technical barriers associated with creating manipulated audiovisual content. Deepfake generation systems can synthesize or manipulate facial identities, expressions, speech, and other visual characteristics with increasing realism. This development has created legitimate applications in entertainment and digital media while simultaneously introducing risks associated with misinformation, impersonation, fraud, and loss of trust in digital evidence.

Deepfake detection has therefore emerged as an important research problem within computer vision and digital media forensics. Early approaches investigated visible manipulation artifacts such as face-warping distortions and physiological inconsistencies. Li and Lyu, for example, demonstrated that geometric artifacts introduced during face warping could provide useful evidence for detecting DeepFake videos [1]. Li et al. subsequently investigated eye-blinking behavior as a physiological cue for identifying synthesized facial videos [2].

The development of larger benchmark datasets enabled researchers to evaluate deepfake detectors more systematically. Rossler et al.

introduced FaceForensics++, which provided a benchmark containing multiple facial manipulation techniques and different compression conditions [3]. The DeepFake Detection Challenge (DFDC) dataset subsequently provided a much larger collection of manipulated videos for training and evaluating detection systems [4]. Celeb-DF was introduced to address limitations associated with the visual quality of earlier datasets and to provide more challenging, high-quality deepfake examples [5].

Deep learning has become a dominant approach in this field. CNN-based systems can learn spatial representations associated with facial texture, boundaries, lighting, and other manipulation artifacts. Wodajo and Atnafu proposed a Convolutional Vision Transformer (CViT), combining CNN-based feature extraction with a Vision Transformer to model visual representations using attention [6]. Their reported evaluation on DFDC achieved 91.5% accuracy and an AUC of 0.91.

However, a video contains more information than independent images. Facial appearance and motion evolve continuously over time. Consequently, a manipulated video may exhibit inconsistencies in facial movement, expression transitions, head motion, lip movement, visual stability, or other temporal characteristics that may not be sufficiently represented by independent frame classification. Cozzolino et al. addressed this direction through ID-Reveal, which learns identity-aware temporal facial features and reported improved robustness to low-quality videos and unseen manipulation types [7].

Multimodal approaches have further demonstrated that deepfake detection can benefit from combining different information sources. Elpeltagy et al. proposed a system combining visual and audio representations with GRU-based attention mechanisms and evaluated it on FakeAVCeleb [8]. Such approaches demonstrate the potential of integrating temporal and multimodal information, although their computational complexity can be substantial.

Another important limitation is interpretability. A classification system that outputs only “REAL” or “FAKE” provides limited information about why the decision was made. This can be problematic when the system is intended to assist forensic analysts or users who require evidence to support a decision. Grad-CAM, introduced by Selvaraju et al., provides a method for visualizing regions that influence CNN predictions [9]. Such techniques can potentially provide a human-interpretable layer for deepfake detection.

The proposed project therefore focuses on an integrated framework called Deepfake Video Authentication using Explainable Temporal AI. Rather than treating authentication as a simple binary classification task, the proposed system combines spatial facial analysis, temporal consistency modeling, explainable prediction, suspicious temporal-segment localization, and calibrated authenticity scoring. The primary implementation will use an Xception spatial backbone and a Temporal Transformer, with LSTM and GRU retained as comparative baselines.

II. RELATED WORK

A. Spatial Artifact-Based Detection

Early deepfake detection research concentrated on visual artifacts produced during facial manipulation. Li and Lyu investigated face-warping artifacts caused by transforming synthesized facial regions to match the geometry of the original face [1]. Such artifacts can provide useful detection cues because image synthesis and geometric transformation may introduce characteristic distortions.

Rossler et al. later introduced FaceForensics++ as a benchmark for facial manipulation detection [3]. The benchmark includes multiple manipulation techniques, including DeepFakes, Face2Face, FaceSwap, and NeuralTextures. Their work demonstrated that data-driven detectors can achieve strong performance, particularly when domain-specific information is incorporated.

The main strength of spatial approaches is their ability to identify localized facial artifacts. However, their representation is primarily based on individual frames. If a manipulated frame does not contain strong visible artifacts, frame-level classification may become difficult.

B. Temporal Deepfake Detection

Temporal analysis attempts to model how facial information changes between consecutive frames. Guera and Delp investigated recurrent neural networks for deepfake video detection, demonstrating the use of sequential modeling for identifying temporal information in manipulated videos [10].

Temporal approaches are motivated by the observation that a deepfake can appear visually plausible in an individual frame while still producing abnormal changes across a sequence. Potential cues include inconsistent expressions, unstable facial landmarks, unnatural head movement, inconsistent lip motion, temporal flickering, and sudden blur or clarity changes.

Cozzolino et al. proposed ID-Reveal, which learns temporal facial characteristics associated with how a particular individual moves while speaking [7]. Rather than requiring fake examples for training, their method learns temporal identity characteristics from real videos and uses these representations to identify deviations.

More recently, Saha et al. investigated the problem of detecting manipulated segments inside otherwise genuine videos [11]. Their work is particularly relevant to the proposed project because it moves beyond video-level binary classification and attempts to identify the temporal intervals containing manipulated content.

C. Transformer-Based Detection

Transformer architectures have increasingly been applied to computer vision because of their ability to model relationships between different portions of an input. Wodajo and Atnafu proposed CViT, combining a CNN feature extractor with a Vision Transformer [6]. Their reported evaluation achieved 91.5% accuracy and 0.91 AUC on DFDC.

M2TR further explored multi-scale transformer representations and frequency-domain information for deepfake detection [12]. These approaches demonstrate that attention mechanisms can provide useful representations for subtle manipulation artifacts.

However, transformer-based spatial approaches do not automatically solve the temporal modeling problem. A model can use attention over spatial patches within an image without explicitly modeling the evolution of facial features across time. The proposed work therefore uses a dedicated Temporal Transformer after spatial feature extraction so that attention operates across facial feature sequences.

D. Multimodal Deepfake Detection

Deepfakes can involve both visual and audio manipulation. Elpeltagy et al. proposed a multimodal system using an upgraded XceptionNet for visual features and a modified InceptionResNetV2-based approach for audio features, followed by GRU-based attention mechanisms [8].

Their results demonstrate the potential of multimodal learning. However, the present project initially concentrates on visual

spatial-temporal analysis so that the temporal and explainability contributions can be evaluated independently. Audio-video fusion is retained as future work.

E. Explainable AI for Deepfake Detection

Deep neural networks are frequently described as difficult to interpret. In a forensic setting, a classification output alone may not provide sufficient evidence for a human analyst. Grad-CAM provides a post-hoc visualization of image regions that contribute to a CNN prediction [9]. Applied to facial deepfake detection, Grad-CAM can highlight areas such as the eyes, mouth, cheeks, forehead, or face boundaries that influenced a prediction.

However, an important distinction must be maintained: a Grad-CAM heatmap is not automatically a ground-truth manipulation mask. It indicates regions that influenced the model's decision and should therefore be treated as an explanatory visualization rather than definitive proof of manipulation.

III. RESEARCH GAP AND PROBLEM FORMULATION

A. Research Gap

The literature demonstrates substantial progress in spatial deepfake detection, temporal modeling, transformer-based representation learning, multimodal analysis, and explainable AI. However, these capabilities are often studied as separate components.

Spatial detectors can identify facial artifacts, but their performance may depend on manipulation-specific visual cues [1], [3], [6]. Temporal approaches explicitly model sequences, but the resulting predictions are not necessarily accompanied by human-interpretable visual evidence [7], [10]. Multimodal systems combine multiple information sources but can require complex architectures and substantial computational resources [8]. Furthermore, conventional deepfake detection commonly produces a video-level decision rather than identifying the temporal intervals that require human inspection.

Temporal segment detection has already been investigated by Saha et al. [11], demonstrating that identifying fake portions within predominantly genuine videos is an important research direction. Therefore, the proposed research does not claim that temporal localization is unexplored. Instead, the specific integration gap addressed in this project is the combination of: Spatial facial analysis + temporal consistency modeling + explainable visual evidence + temporal suspicion localization + calibrated authenticity scoring within a single authentication-oriented workflow.

The research therefore focuses on integration and interpretability rather than claiming that each individual component is novel.

B. Problem Statement

Existing deepfake detection systems can provide high classification performance under particular dataset and manipulation conditions, but a video-level REAL/FAKE prediction may provide insufficient evidence for forensic interpretation. The problem addressed in this research is therefore:

<i>How can spatial facial artifacts and temporal facial inconsistencies be jointly modeled to detect deepfake videos while providing interpretable visual evidence and identifying suspicious temporal segments?</i>

C. Research Questions

- RQ1: Does temporal modeling improve deepfake detection compared with a spatial-only frame-based baseline?
- RQ2: Which temporal architecture—LSTM, GRU, or Temporal Transformer—provides the most effective balance between detection performance and computational complexity?
- RQ3: Can explainability techniques identify facial regions and temporal positions that meaningfully contribute to deepfake

predictions?

- RQ4: Can frame- or sequence-level prediction scores be used to identify suspicious temporal segments within a video?
- RQ5: Does combining spatial and temporal information improve generalization compared with a spatial-only detector?

IV. PROPOSED CONTRIBUTIONS

- Spatial-temporal deepfake authentication framework: A unified pipeline combining spatial facial representations with temporal sequence modeling.
- Temporal facial consistency analysis: Investigation of temporal changes in facial appearance, motion, and visual stability across consecutive video frames.
- Explainable deepfake predictions: Integration of Grad-CAM and temporal attention visualization to provide interpretable evidence associated with model decisions.
- Temporal suspicion localization: Generation of suspicious timestamps or temporal intervals based on sequence-level prediction scores.
- Visual instability and clarity evidence: Reporting model-derived changes in blur/clarity and motion stability as supporting temporal evidence, without interpreting them as proof of manipulation.
- Interpretable authenticity score: Conversion of calibrated model probabilities into an authentication-oriented score.
- Ablation-based evaluation: Comparison between spatial-only, recurrent temporal, and spatial-temporal configurations.

These contributions should be described as proposed contributions until they are supported by experimental results.

V. PROPOSED SYSTEM ARCHITECTURE

The conceptual architecture of the proposed system is shown below:



Fig. 1. Overall architecture and operational workflow.

LSTM and GRU are evaluated as temporal baselines, while the Temporal Transformer is the primary temporal architecture of the proposed framework. The architecture intentionally separates classification from explanation. The classifier determines the estimated probability of manipulation, while the XAI component provides post-hoc evidence about the model's decision.

VI. METHODOLOGY

The proposed methodology consists of video preprocessing, face detection and alignment, facial sequence construction, spatial feature extraction, temporal feature learning, feature fusion, classification, temporal suspicion analysis, and explainability.

A. Video Preprocessing

Given an input video V , the video is decoded into a sequence of frames:

$$V = \{F_{1</sub>}, F_{2</sub>}, \dots, F_{T</sub>}\} \quad (1)$$

where T represents the number of sampled frames. Instead of processing every frame, temporal sampling can be applied at a predefined frame interval. This reduces computational cost while retaining sufficient temporal information. Each selected frame is resized to the input resolution required by the spatial feature extractor. Pixel values are normalized according to the selected backbone's preprocessing requirements.

Candidate parameters include:

- Frame sampling rate: 10 fps
- Sequence length: 16
- Image resolution: 224 × 224
- Batch size: 16

These values must be determined experimentally rather than assumed.

B. Face Detection and Alignment

Deepfake manipulation frequently targets facial regions. Therefore, facial regions can be extracted before spatial feature learning. Candidate detectors include RetinaFace and MTCNN. For each frame, the detected face is cropped and aligned according to facial landmarks. Frames where no reliable face is detected can either be discarded or processed using an appropriate fallback strategy.

For videos containing multiple faces, the system must define a consistent identity-selection strategy, such as: largest detected face, central face, tracked face, or all detected faces. The chosen strategy should be documented because it can influence the experimental results.

VII. FACIAL SEQUENCE CONSTRUCTION

After face extraction, frames are grouped into fixed-length sequences:

$$S_{i</sub>} = \{X_{i</sub>1</sub>}, X_{i</sub>2</sub>}, \dots, X_{i</sub>L</sub>}\} \quad (2)$$

where L is the sequence length and $X_{i</sub>t</sub>}$ represents the facial representation at time t . Sequence construction enables the temporal model to learn relationships between neighboring observations rather than treating frames independently.

VIII. SPATIAL FEATURE EXTRACTION

The spatial component extracts visual information from individual facial frames. The primary proposed backbone is Xception because CNN-based deepfake detectors have demonstrated effectiveness on facial manipulation benchmarks [3], [4]. EfficientNet may be retained as a comparative spatial architecture where resources permit.

The spatial feature extractor transforms each facial frame into:

$$z_{t</sub>} = f_{0</sub>}(X_{t</sub>}) \quad (3)$$

where $f_{0</sub>}$ represents the CNN backbone and $z_{t</sub>}$ is the resulting feature representation. These representations may contain information related to facial texture, skin consistency, blending boundaries, lighting, local facial structure, and compression artifacts. The spatial model should not be assumed to identify manipulation-specific pixels perfectly. Its learned features are statistical representations used by the classifier.

IX. TEMPORAL FEATURE LEARNING

The primary research focus is temporal modeling. The sequence of spatial features is represented as:

$$Z = \{z_{1</sub>}, z_{2</sub>}, \dots, z_{L</sub>}\} \quad (4)$$

A temporal model $g_{\phi</sub>}$ processes this sequence:

$$h_{1</sub>}, \dots, h_{L</sub>} = g_{\phi</sub>}(z_{1</sub>}, \dots, z_{L</sub>}) \quad (5)$$

The primary temporal architecture is a Temporal Transformer. LSTM and GRU are retained as recurrent comparison models.

A. LSTM

Long Short-Term Memory (LSTM) can model long-term dependencies through gated memory mechanisms and provides a recurrent baseline for temporal sequence learning [10].

B. GRU

Gated Recurrent Unit (GRU) provides recurrent sequence modeling with fewer gating components than LSTM and can provide a useful comparison for sequence learning [8].

C. Temporal Transformer

A Temporal Transformer models relationships between different temporal positions using self-attention. In the proposed framework, attention is applied to the sequence of facial feature representations rather than only to spatial patches within a single image. This allows the model to learn relationships between facial states across time and also provides temporal attention weights that can be visualized as supporting evidence. Transformer-based attention has been applied to deepfake detection in prior work [6], [12].

X. SPATIAL-TEMPORAL FEATURE FUSION

The spatial and temporal representations can be combined before classification. A simple fusion strategy is:

$$h_{\text{fusion}} = [z_{\text{spatial}}; h_{\text{temporal}}] \quad (6)$$

where $[\cdot]$ represents concatenation. A fully connected layer can then transform the fused representation before the final classification layer. The purpose of fusion is to allow the classifier to use both what the face looks like and how the face changes over time.

XI. DEEPPAKE CLASSIFICATION

The final classification layer produces:

$$P(\text{fake} | V) \quad (7)$$

where the output represents the estimated probability that the input video is manipulated. A threshold τ can be selected using the validation set:

$$\blacksquare = \text{Fake, if } P(\text{fake} | V) \geq \tau; \text{ Real, if } P(\text{fake} | V) < \tau \quad (8)$$

The threshold should not automatically be assumed to be 0.5. It should be selected according to the evaluation objective and validation performance. Binary cross-entropy may be used for training:

$$L = -[y \log(p) + (1 - y) \log(1 - p)] \quad (9)$$

where y is the ground-truth label and p is the predicted fake probability.

XII. TEMPORAL SUSPICION LOCALIZATION

A major objective of the proposed system is to move beyond a single video-level prediction. Suppose the video is divided into N temporal windows:

$$W_1, W_2, \dots, W_N \quad (10)$$

The model produces a fake probability for each window:

$$p_1, p_2, \dots, p_N \quad (11)$$

A temporal suspicion curve can then be generated. For example, Table I illustrates the intended output format.

TABLE I
EXAMPLE TEMPORAL SUSPICION ANALYSIS

Segment	Fake Probability	Interpretation
00:00–00:03		Low suspicion
00:03–00:05		High suspicion
00:05–00:08		Moderate suspicion
00:08–00:10		High suspicion

A smoothing operation may be applied to reduce isolated noisy predictions. Adjacent high-probability windows can be merged into larger suspicious segments. For example:

$$[00:03, 00:04] + [00:04, 00:05] \rightarrow [00:03, 00:05] \quad (12)$$

This produces a timeline that can help a human reviewer focus on potentially manipulated portions. Importantly, if the dataset does not provide ground-truth manipulated intervals, these timestamps should be described as model-derived suspicious intervals, not verified manipulation boundaries.

XIII. EXPLAINABLE ARTIFICIAL INTELLIGENCE

A. Grad-CAM

Grad-CAM can be applied to the CNN component to generate a spatial heatmap. The method uses gradients flowing toward a convolutional layer to determine which regions contribute to a target prediction [9]. The output can be represented as:

$$\text{Original Face} \rightarrow \text{Grad-CAM Heatmap} \rightarrow \text{Overlay}$$

Potentially highlighted areas may include the eyes, mouth, cheeks, forehead, face boundary, and nose region. However, the heatmap should be interpreted carefully. A highlighted region means “This region influenced the model’s prediction.” It does not necessarily mean “This exact region contains the ground-truth manipulation.” This distinction is essential for forensic credibility.

B. Temporal Attention Visualization

Because the primary proposed temporal architecture is a Transformer, temporal attention weights can be visualized to identify sequence positions receiving greater model attention. This complements Grad-CAM by providing evidence about when the model relied more strongly on temporal information.

TABLE II
EXAMPLE TEMPORAL ATTENTION VISUALIZATION

Frame/Time	Relative Attention
00:03.0	
00:03.5	
00:04.0	
00:04.5	

Attention visualization can complement, but should not automatically be treated as a definitive explanation of model causality.

XIV. AUTHENTICITY AND TRUST SCORE

The proposed system can produce a calibrated authenticity score rather than an arbitrary score. If the calibrated probability of manipulation is $P_{\text{fake}}(V)$, then an authenticity score can be defined as:

$$A(V) = 100 (1 - P_{\text{fake}}(V)) \quad (13)$$

where $A(V) \approx 100$ indicates high estimated authenticity and $A(V) \approx 0$ indicates high estimated manipulation probability. However, the probability should preferably be calibrated using a validation set before being presented as a trust-oriented score. Therefore, the dashboard should distinguish between Model Confidence and Authenticity Score rather than implying that the system has established absolute truth. A more responsible interpretation is: “Estimated authenticity based on the trained model and evaluation conditions.”

XV. DATASETS

The following datasets are appropriate candidates.

A. FaceForensics++

FaceForensics++ provides manipulated facial videos generated using several manipulation techniques and different compression conditions [3]. It is useful for evaluating spatial and cross-manipulation performance.

A. DFDC

The DeepFake Detection Challenge dataset provides a large collection of face-swapped videos and was specifically designed to support deepfake detection research [4].

B. Celeb-DF

Celeb-DF contains high-quality deepfake videos and was designed to address some visual-quality limitations of earlier datasets [5].

C. FakeAVCeleb

FakeAVCeleb is particularly relevant if the project later incorporates audio-video multimodal detection. Elpeltagy et al. used this dataset for their multimodal deepfake detection system [8].

D. Temporal Segment Dataset

If temporal localization becomes a major evaluation objective, the benchmark introduced by Saha et al. is particularly relevant because it contains videos with real and manipulated temporal portions [11].

XVI. EXPERIMENTAL SETUP

The experiments should be designed to isolate the contribution of temporal modeling.

A. Experiment A: Spatial Baseline

Input → Face Detection → Xception/EfficientNet → Classifier

Purpose: Establish the performance of a conventional spatial detector.

B. Experiment B: CNN + LSTM

Input → CNN → LSTM → Classifier

Purpose: Determine whether recurrent temporal sequence modeling improves upon the spatial baseline.

C. Experiment C: CNN + GRU

Input → CNN → GRU → Classifier

Purpose: Evaluate whether GRU provides a useful temporal comparison against LSTM.

D. Experiment D: CNN + Temporal Transformer

Input → CNN → Temporal Transformer → Classifier

Purpose: Evaluate attention-based temporal modeling across face sequences.

E. Experiment E: Proposed Spatial-Temporal Framework

Input → Xception → Temporal Transformer → Fusion → Classifier → XAI → Temporal Localization → Authenticity Score

Purpose: Evaluate the complete proposed workflow.

XVII. ABLATION STUDY

Ablation experiments should be performed to determine which components contribute to performance. XAI itself should not be evaluated as a classification component unless it is integrated into the model architecture. Instead, its usefulness and faithfulness should be assessed separately.

TABLE III
ABLATION STUDY CONFIGURATION

Model Configuration	Spatial	Temporal	XAI	Temporal Localization	Result
Baseline CNN	Yes	No	No	No	
CNN + LSTM	Yes	Yes	No	No	
CNN + GRU	Yes	Yes	No	No	
CNN + Transformer	Yes	Yes	No	No	
Proposed Framework	Yes	Yes	Yes	Yes	

XVIII. EVALUATION METRICS

A. Accuracy

$$Accuracy = (TP + TN) / (TP + TN + FP + FN) \quad (15)$$

Accuracy measures the overall percentage of correctly classified samples.

B. Precision

$$Precision = TP / (TP + FP) \quad (16)$$

Precision measures how many samples predicted as fake are actually fake.

C. Recall

$$Recall = TP / (TP + FN) \quad (17)$$

Recall measures how many actual fake videos are successfully detected.

D. F1-Score

$$F1 = 2 \times Precision \times Recall / (Precision + Recall) \quad (18)$$

F1-score balances precision and recall.

E. ROC-AUC

ROC-AUC evaluates discrimination across different classification thresholds. This is particularly useful when the class distribution is not perfectly balanced.

F. Temporal Localization Metrics

If ground-truth manipulated intervals are available, the project should evaluate temporal Intersection-over-Union (tIoU), temporal precision, temporal recall, and segment-level F1-score. If ground-truth temporal annotations are unavailable, the project should not claim quantitative temporal localization accuracy.

XIX. EXPECTED RESULTS

The following table should be completed only after experiments. No numerical result is assumed in this paper.

TABLE IV
EXPECTED EXPERIMENTAL RESULTS

Model	Dataset	Accuracy	Precision	Recall	F1	ROC-AUC
Baseline CNN/Xception	FaceForensics++	88.4	87.9	86.8	[RESULT]	87.3
CNN + LSTM	FaceForensics++	92.1	90.1	89.7	[RESULT]	89.2
CNN + GRU	FaceForensics++	89.4	94.0	90.6	[RESULT]	90.2
CNN + Transformer	FaceForensics++	89.9	90.0	94.4	[RESULT]	92.3
Proposed Framework	FaceForensics++	92.0	91.5	91.7	[RESULT]	96.1

Note: Numerical values shown here are illustrative target values for planning/layout only; they are not measured experimental results.

XX. COMPARATIVE ANALYSIS WITH EXISTING METHODS

Table V summarizes representative existing methods and their relationship to the proposed research. The Author/Year field is paired with the corresponding citation number so that each comparison can be traced directly to the bibliography.

TABLE V
COMPARISON WITH EXISTING METHODS

Author/Year [Citation]	Method/Architecture	Dataset	Main Contribution	Limitation Relevant to Proposed Work
Li and Lyu (2018) [1]	CNN-based face-warping artifact detection	DeepFake datasets	Detects artifacts produced by face warping	Primarily dependent on spatial manipulation artifacts.
Li et al. (2018) [2]	Eye-blinking analysis	DeepFake videos	Uses physiological behavior as a detection cue	Biological cue may be less reliable in modern high-quality videos.
Rossler et al. (2019) [3]	CNN-based facial manipulation detection / FaceForensics++	FaceForensics++	Established a major benchmark for facial manipulation detection	Benchmark-oriented detection does not itself provide temporal suspicion localization.

Author/Year [Citation]	Method/Architecture	Dataset	Main Contribution	Limitation Relevant to Proposed Work
Dolhansky et al. (2020) [4]	DFDC dataset and benchmark	DFDC	Large-scale dataset for deepfake detection	Dataset is a benchmark resource rather than an explainable authentication framework.
Li et al. (2020) [5]	Celeb-DF dataset	Celeb-DF	Provides high-quality and challenging deepfake videos	Primarily a dataset contribution.
Wodajo and Atnafu (2021) [6]	CNN + Vision Transformer	DFDC	Combines CNN feature extraction with Transformer attention; reported 91.5% accuracy and 0.91 AUC	Primarily focuses on spatial representation and does not provide the proposed integrated temporal-XAI workflow.
Cozzolino et al. (2021) [7]	ID-Reveal; identity-aware temporal modeling	Multiple benchmarks	Learns temporal facial behavior and improves generalization	Requires identity/reference information and does not constitute the proposed XAI + temporal-localization workflow.
Elpeltagy et al. (2023) [8]	XceptionNet + InceptionResNetV2 + GRU attention	FakeAVCeleb	Combines visual, audio, and temporal information	Computationally complex; multimodal focus differs from the proposed visual explainability-centered framework.
Guera and Delp (2018) [10]	CNN + recurrent neural network	Deepfake videos	Investigates recurrent temporal modeling	Does not provide the proposed unified explainability and temporal-localization workflow.
Wang et al. (2022) [12]	M2TR multi-scale Transformer	Deepfake benchmarks	Models multi-scale and frequency-domain forgery information	Focuses mainly on manipulation representation rather than temporal suspicion explanation.
Saha et al. (2023) [11]	Spatial + Time-Series Transformer	Temporal deepfake segment benchmark	Detects manipulated temporal segments rather than only whole-video labels	Does not specifically integrate the proposed Grad-CAM-based spatial explanation and authenticity-score workflow.
Yan et al. (2023) [13]	DeepfakeBench	Multiple datasets	Standardizes datasets, preprocessing, detectors, and evaluation	Benchmark framework rather than an explainable temporal authentication model.
Kaur et al. (2024) [14]	Survey of deepfake video detection	Multiple studies	Highlights temporal continuity as a major challenge/opportunity	Survey-based; does not provide the proposed unified implementation.

XXI. DISCUSSION OF THE RESEARCH GAP

The comparative analysis indicates that existing research has addressed individual components of the proposed problem. Spatial artifact methods provide useful visual evidence [1], [3]. Transformer-based approaches improve representation learning [6], [12]. Temporal methods demonstrate that facial behavior and frame continuity can provide additional information [7], [10], [11]. Multimodal systems demonstrate the value of combining different information sources [8]. Benchmark studies emphasize the importance of standardized evaluation and generalization [4], [13].

However, these capabilities are not necessarily provided together in a single practical and interpretable authentication workflow. The proposed project therefore investigates an integrated approach in which:

- spatial features identify facial visual evidence;
- temporal modeling analyzes facial consistency and motion evolution;
- classification estimates manipulation probability;
- temporal scores identify suspicious intervals;
- Grad-CAM provides spatial explanations;
- temporal attention visualization provides supporting temporal evidence;
- blur/clarity and visual-stability changes may be reported as model-derived evidence where measurable;
- calibrated probability is converted into an interpretable authenticity score.

The novelty should therefore be positioned as system-level methodological integration and forensic interpretability, rather than claiming novelty for CNNs, LSTMs, Transformers, Grad-CAM, or temporal localization individually.

XXII. EXPLAINABILITY AND FORENSIC INTERPRETATION

A central objective of this project is to make the detection process more interpretable. A conventional system might produce:

Prediction: FAKE — 94.2%

The proposed system aims to provide a richer output:

- Prediction: FAKE
- Model probability: 0.94
- Authenticity score: 94
- Suspicious interval: 00:03–00:05
- Dominant visual evidence: mouth and lower-face region
- Temporal evidence: abnormal feature variation across consecutive frames
- Visual stability evidence: moderate variation, where measurable

This output is intended to assist human inspection. However, the system should not claim: “The mouth is definitely manipulated.” Instead, it should report: “The mouth region contributed strongly to the model’s prediction.” This distinction is especially important because post-hoc explanations such as Grad-CAM are not equivalent to ground-truth manipulation annotations [9].

XXIII. SYSTEM OUTPUT AND AUTHENTICATION DASHBOARD

The proposed dashboard should contain the following components.

A. Video Information

- Uploaded video name
- Duration
- Resolution
- Frame rate

B. Authentication Result

- REAL / FAKE
- Model probability
- Calibrated authenticity score

C. Temporal Evidence

A timeline can be represented as: 00:00 — 00:03 — 00:05 — 00:08 — 00:10 with suspicious intervals highlighted according to model-derived scores.

D. Spatial Explanation

- Original frame
- Grad-CAM visualization
- Overlay visualization

E. Temporal Analysis

- Sequence probability curve
- Suspicion peaks
- Temporal Transformer attention visualization
- Motion/visual-stability indicators
- Blur/clarity variation indicators, where supported by the analysis

The dashboard should distinguish model evidence from verified ground truth. Suspicious intervals and highlighted regions are model-derived evidence unless validated against suitable annotations.

XXIV. LIMITATIONS

Several limitations should be acknowledged. First, deepfake datasets may contain biases associated with identities, recording environments, manipulation methods, and compression conditions. A model may therefore learn dataset-specific characteristics rather than general forgery properties. Second, performance on one benchmark does not guarantee performance on unseen deepfake generation methods. Cross-dataset testing should therefore be included where computationally feasible. Third, compressed or low-resolution videos can remove both genuine and manipulated visual signals. Fourth, multiple faces, occlusions, extreme head poses, and rapid movements may make consistent facial tracking difficult. Fifth, temporal localization requires suitable temporal ground truth for quantitative evaluation. If the selected dataset contains only video-level labels, the resulting suspicious timestamps are model-derived indicators rather than validated manipulation boundaries. Sixth, Grad-CAM and attention visualizations should not be interpreted as exact manipulation masks. Seventh, blur/clarity changes and motion instability can arise from genuine recording conditions and therefore must not be treated as manipulation proof by themselves. Finally, Temporal Transformers and large CNN backbones may require substantial computational resources. Model complexity should therefore be considered alongside accuracy.

XXV. FUTURE WORK

Future work can extend the framework in several directions. First, audio-video multimodal analysis could be incorporated to detect inconsistencies between speech, lip movement, and facial behavior. Multimodal approaches such as the system proposed by Elpeltagy et al. demonstrate the feasibility of this direction [8]. Second, larger temporal Transformer architectures could be investigated for longer-range facial behavior modeling. Third, cross-dataset evaluation could be expanded to measure robustness against unseen manipulation techniques. Fourth, real-time inference could be investigated through efficient temporal sampling, model quantization, and knowledge distillation. Fifth, temporal localization could be improved using datasets containing frame-level or segment-level manipulation annotations. Sixth, explanation evaluation could be extended beyond visual inspection by studying explanation faithfulness, stability, and agreement with available manipulation annotations. Finally, future systems could investigate multimodal forensic evidence including visual artifacts, temporal inconsistencies, audio anomalies, lip-sync relationships, physiological signals, and compression characteristics.

XXVI. CONCLUSION

This paper proposed an explainable spatial-temporal framework for deepfake video authentication. Existing research demonstrates strong capabilities in spatial artifact detection, temporal modeling, Transformer-based representation learning, multimodal detection, and benchmark development. Nevertheless, a practical authentication system requires more than a single video-level classification.

The proposed framework therefore investigates the integration of Xception-based spatial facial analysis, Temporal Transformer-based consistency modeling, explainable AI, temporal suspicion localization, and calibrated authenticity scoring. The primary research focus is temporal modeling, with the objective of determining whether analyzing facial sequences provides useful information beyond independent frame-level analysis.

The proposed experimental design compares spatial-only, recurrent temporal, and Transformer-based temporal models. Explainability is incorporated through Grad-CAM and temporal attention visualization. Suspicious temporal segments are generated from sequence-level prediction scores, while the authenticity score is derived from calibrated manipulation probability. Visual stability and blur/clarity changes are treated as supporting model-derived evidence rather than definitive indicators of manipulation.

The final scientific contribution will depend on the experimental findings. In particular, the project should determine whether the proposed integration provides measurable improvements in detection performance, temporal localization, interpretability, or practical forensic usability. Consequently, no numerical performance claims are made for the proposed system until experiments are completed.

XXVII. RESEARCH QUESTIONS AND HYPOTHESES

A. Research Questions

- RQ1: Does incorporating temporal information improve deepfake detection compared with spatial-only classification?
- RQ2: Which temporal architecture provides the best balance between detection performance and computational cost?
- RQ3: Can Grad-CAM and temporal attention visualizations provide useful evidence for interpreting deepfake predictions?
- RQ4: Can model-derived temporal scores identify suspicious portions of manipulated videos?
- RQ5: Does the proposed spatial-temporal framework generalize better across datasets than a spatial-only baseline?

B. Hypotheses

- H1: A spatial-temporal model will achieve higher F1-score and ROC-AUC than a spatial-only baseline under the same experimental protocol.
- H2: Temporal modeling will improve detection of manipulations that exhibit inconsistencies across consecutive frames.
- H3: Explainability visualizations will provide meaningful evidence about the facial regions and temporal positions influencing model predictions, although they will not necessarily correspond exactly to manipulation masks.
- H4: Sequence-level prediction scores can be used to identify suspicious temporal intervals when suitable temporal ground truth is available.

XXVIII. NOVELTY AND UNIQUE SELLING PROPOSITION

The proposed work should be positioned as follows:

<i>The novelty of the proposed framework lies not in introducing a new CNN, recurrent network, Transformer, or XAI algorithm individually, but in integrating spatial deepfake analysis, temporal facial consistency modeling through a dedicated Temporal Transformer, explainable prediction, temporal suspicion localization,

visual-stability/clarity evidence, and calibrated authenticity scoring into a unified video-authentication workflow.</i>

This positioning is academically safer because related literature has already explored temporal modeling, explainability, multimodal analysis, and temporal segment detection individually [7]–[9], [11]. The project’s USP is therefore the unified forensic workflow that connects what the model detects, when suspicious behavior occurs, and which evidence influenced the prediction, while explicitly distinguishing model-derived evidence from verified manipulation ground truth.

XXIX. LIST OF EXPERIMENTS TO PERFORM

- 1) Train and evaluate a CNN/Xception spatial baseline.
- 2) Train CNN + LSTM.
- 3) Train CNN + GRU.
- 4) Train CNN + Temporal Transformer as the primary temporal architecture.
- 5) Compare all models using identical train/validation/test protocols.
- 6) Perform cross-dataset evaluation where feasible.
- 7) Evaluate robustness under video compression.
- 8) Generate Grad-CAM visualizations.
- 9) Generate temporal attention visualizations for the Transformer.
- 10) Generate frame/sequence-level fake probabilities.
- 11) Generate temporal suspicion curves.
- 12) Evaluate temporal localization if ground-truth intervals are available.
- 13) Evaluate model-derived blur/clarity and visual-stability indicators without treating them as definitive manipulation labels.
- 14) Calibrate model probabilities.
- 15) Generate the final authenticity score.
- 16) Perform an ablation study.

XXX. INFORMATION REQUIRED BEFORE FINAL PUBLICATION

[To be completed after implementation: final dataset split, preprocessing parameters, hardware configuration, training settings, experimental results, confusion matrices, ROC curves, temporal localization results, explanation examples, and statistical comparisons.]

XXXI. RECOMMENDED FIGURES

The following figures are recommended for the final implementation.

- 1) Fig. 1. Overall architecture of the proposed explainable temporal deepfake authentication framework.
- 2) Fig. 2. Video preprocessing, frame sampling, face detection, and facial sequence construction pipeline.
- 3) Fig. 3. Xception spatial feature extraction and Temporal Transformer feature learning architecture.
- 4) Fig. 4. Spatial-temporal feature fusion and deepfake classification pipeline.
- 5) Fig. 5. Grad-CAM-based explanation of suspicious facial regions.
- 6) Fig. 6. Temporal suspicion probability curve and suspicious-segment localization.
- 7) Fig. 7. Example authentication dashboard showing classification, authenticity score, temporal evidence, and visual explanation.
- 8) Fig. 8. Comparative performance of spatial-only, recurrent, Transformer-based, and proposed models.

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